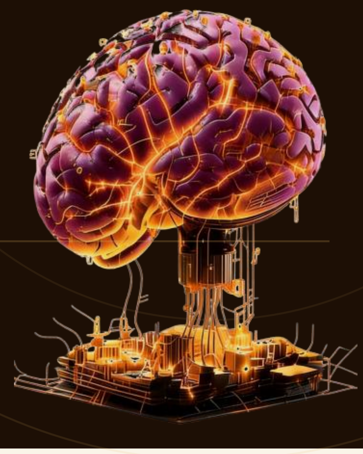
DEEP MINDS · 7TH EDITION

Special PROJECT SHOWCASE

Drone Detection System

A real time computer vision system for detecting, segmenting, and tracking unmanned aerial vehicles using **YOLO** and ByteTrack.



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PROJECT CODE

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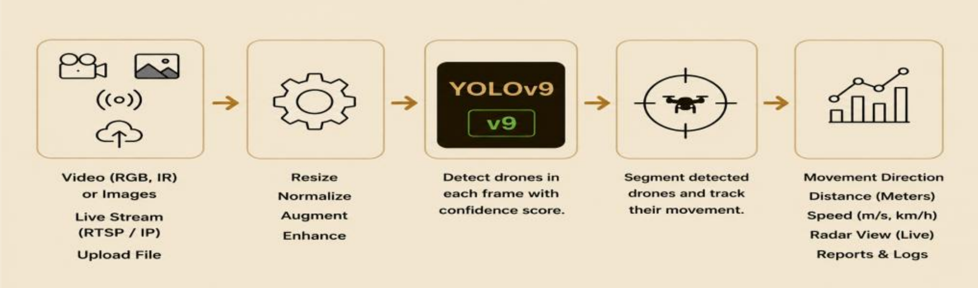
Abstract

This project presents an AI powered drone detection system for real time UAV detection, segmentation, and tracking. Built using a custom trained YOLO model, ByteTrack, and OpenCV, the system provides accurate detection, efficient processing, and a scalable architecture

Methodology



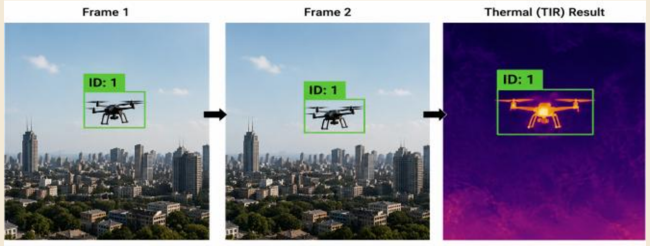
Results & Evaluation

98.4%	0.98	10ms	250+
Accuracy	mAP@0.5	Avg. Inference Time	Test Videos

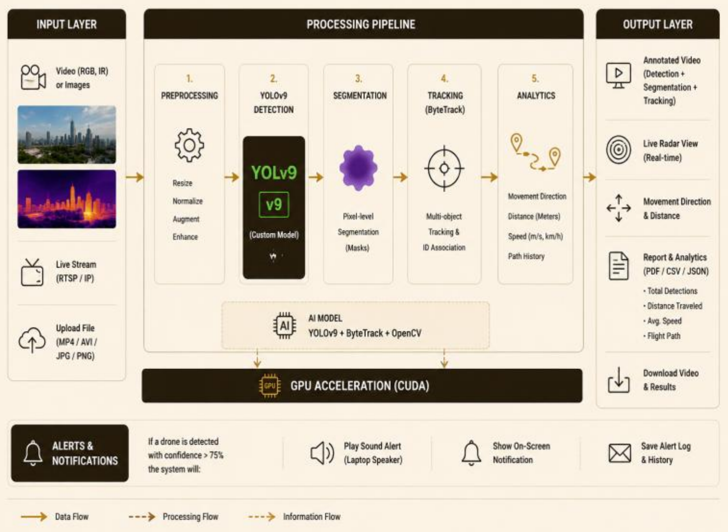
Dataset

Anti **UAV RGBT** benchmark containing visible and thermal drone data with separate training, Validation, and testing sets.

57.4K	1	50/20/30
SAMPLES	CLASSES	SPLIT



System Architecture



Use Case

Conclusion

The proposed system demonstrates accurate and efficient real time drone detection, segmentation, and tracking using **YOLOv9** and **ByteTrack**. With live monitoring, movement analysis, and automated alerts, it provides a reliable foundation for future intelligent UAV surveillance applications.

Future Work

- **Train advanced models** on a larger and more diverse dataset to improve detection accuracy in complex environments.
- **Deploy on edge devices** for real-time monitoring with low latency in resource-constrained environments.
- **Integrate multi-sensor data** such as radar and acoustic sensors to enhance detection robustness and reduce false alarms.
- **Improve tracking algorithms** to handle dense drone scenarios and occlusions more effectively.
- **Develop cloud-based platform** for centralized monitoring, data storage, and analytics dashboard.

DEMO & PRESENTATION

Scan to view the live demo and full project report.

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